

GréineGrid Qagent — 5-Minute Defence: Speaker Notes

B9AI105 CA1 · Deck: GreineGrid_RL_CA1_Defence_v2.pptx (8 slides) · 5 min timed + 15 min Q&A

Before you start (all 3)

- Rehearse the full run at 4:45 — 15-second buffer before the hard 5:00 cut.
- Do not read slides. Slides are visual prompts; speak the story.
- Handoffs by name: “I’ll hand to [Name] for...”
- Demo (slides 1, 6, 8): mention the URL only — no live walkthrough unless the panel asks. Verify the deployment is up on the morning of the defence; fall back to local ``streamlit run dashboard/app.py``.
- **Every member must be able to write the Q-Learning update on the board and explain the reward function unaided — the Q&A exists to confirm ownership.**

Timing summary

Member	Slides	Time	Cumulative
Member 1	1–3	1:25	1:25
Member 2	4–5	1:20	2:45
Member 3	6–7	1:10	3:55
Member 1 (close)	8	0:25	4:20 + transitions ≈ 4:45

Slide-by-slide

Slide 1 — Title

Presenter: Member 1 · Time: 0:15

Say: “We’re GréineQ — reinforcement learning for home battery dispatch on a solar microgrid. We built tabular Q-Learning and SARSA from scratch in NumPy, on real Ausgrid and AEMO data. Live demo at greineq-agent.sudocod.com.”

Don’t: Read the subtitle or explain the branding. 15 seconds, move on.

Slide 2 — The problem

Presenter: Member 1 · Time: 0:35

Say: “Solar peaks at midday; demand peaks morning and evening — point at the chart. Without a battery, surplus is exported cheap and the evening is bought dear. The controller decides every 30 minutes: hold, charge, or discharge — 48 steps a day. This is sequential decision-making, not prediction: today’s action changes tomorrow’s battery state.”

Point at: Daily profile chart (midday PV bell vs evening load/price peak).

Slide 3 — Method in three numbers

Presenter: Member 1 · Time: 0:35

Say: “We model one day as a 48-step MDP: 324 discretised states — state of charge, solar, load, price, time of day — and three actions. Physics: 10 kWh battery, 5 kW limits, SOC

clamped 10–90%. Q-Learning and SARSA implemented from scratch in NumPy, compared against four baselines from a no-battery floor to a perfect-foresight oracle.”

Handoff: “[Member 2] will take reward design and results.”

Slide 4 — Reward redesign (strongest argument)

Presenter: Member 2 · Time: 0:40

Say: “Our first reward was a wholesale-price proxy. The agent looked good in training and lost on the real bill — reward–metric misalignment. v2 monetises everything in AUD: retail margin on imports, feed-in value of wasted solar, optional degradation. Now reward ranking matches bill ranking — and we verified that in an ablation.”

Key phrase: “The training reward diverged from the evaluation metric.” Treat this slide as the project’s core contribution, not a detail.

Slide 5 — Main results

Presenter: Member 2 · Time: 0:40

Say: “No battery: 160.75 AUD over 53 days. The perfect-foresight oracle: 49.19 — the theoretical floor. Q-Learning 62.02 and SARSA 63.75 beat the tertile rule by 31–33%. But greedy self-consumption reaches 52.15. RL beats the weak rule, not the strong heuristic — and we report that honestly.”

Don’t: Hide that greedy wins — it sets up the next slide, which is the headline.

Handoff: “[Member 3] will explain why that matters.”

Slide 6 — Headline finding — 97.4%

Presenter: Member 3 · Time: 0:35

Say: “The most important result is not that RL beats every baseline — it’s that a simple heuristic already captures 97.4% of what’s achievable. With solar-only charging and coarse states, using every surplus and covering every deficit is near-optimal.”

Visual: Digital-twin screenshot; open the live dashboard only if the panel asks.

Slide 7 — What we’re not claiming

Presenter: Member 3 · Time: 0:35

Say: “Known limits: one customer, solar-only dispatch, coarse time bins that break the strict Markov property. 5 seeds × 10k episodes: Q-Learning 61.75 ± 2.14 AUD, SARSA 61.62 ± 0.86 AUD, paired t-test $p = 0.90$. RL’s value rises with grid charging, price arbitrage, and fleet coordination — scoped for CA2 on this same MDP and codebase.”

Memorise (all 3): “RL value increases with grid charging, export tariffs, or fleet-level aggregation — flagged as CA2 extensions, not because RL failed here.”

Slide 8 — Conclusion

Presenter: Member 1 (who opened, closes) · Time: 0:25

Say: Four takeaways, spoken to the panel, not the slide: reward design determines what the agent learns; RL beats weak rules, not strong heuristics; greedy sets a realistic ceiling at 97.4%; CA2 adds grid charging and arbitrage. End with “Questions?” — then stop talking. Let the silence work.

15-minute Q&A — who answers what

Member 1 — problem, data, MDP

- Why RL vs supervised / rules? → no labels for optimal actions; rules are myopic; sequential structure (SOC carries value forward).

- Why 324 states? → tertile bins × 4 time bins keep an exact, interpretable table; one more bin per feature multiplies the table ~81×
- Data pipeline? → Ausgrid GC/GG unpivoted, AEMO NSW1 merged on timezone-aligned timestamps; month-stratified 256/54/53 split, seed 50; DST days dropped.

Member 2 — algorithms, reward, numbers

- Q-Learning update on the board: $Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$; SARSA replaces max with the action actually taken.
- Why v1 reward failed → proxy diverged from the bill; agents optimised the wrong objective (Table VI).
- Multi-seed: Q-Learning 61.75 ± 2.14 , SARSA 61.62 ± 0.86 , $p = 0.90$ — statistically tied; bottleneck is problem structure, not algorithm.
- Why RL doesn't beat greedy → coarse bins + solar-only charging; greedy captures 97.4% of the oracle gap.

Member 3 — discussion, demo, limitations

- Why not DQN/PPO? → 324 states: tabular is exact and interpretable; deep RL is for CA2 if the state space grows.
- Double Q-Learning → implemented to reduce maximisation bias; a terminal-update bug was found and fixed; not fully retrained — say this before they ask.
- Dashboard walkthrough; CA2 plans; ethics/limitations (2012–13 data vs 2026 tariffs; simulation, not deployment advice).

AI usage (any member)

- **Level 4 declared. Answer pattern:** “We used [tool] for [task]; I verified it by [test]; one thing it got wrong was [example], which we fixed by [fix].” Keep it specific, not defensive.

Emergency cuts (if running long)

- Slide 3 detail → one sentence: “48-step daily MDP, 324 states, 3 actions, from scratch.”
- Slide 6 KPIs → say only the 97.4% line.
- Always keep slides 4, 5, and 7 intact — they carry the marks (reward design, results, honest limitations).